$$\begin{bmatrix} 5 & 7 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[-7\times Row2]{Row1 = Row1} \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \xrightarrow[\times 1/5]{Row1 = Row1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

From Example 2, we know the final matrix is in rref. There are pivots in all columns of rref of A. By defn, there are no free variables. By Theorem 2, the system has only a unique solution (the trivial solution). ∴ By defn, the vectors $\begin{bmatrix} 5 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 7 \\ 2 \\ -6 \end{bmatrix}$ and $\begin{bmatrix} 9 \\ 4 \\ -8 \end{bmatrix}$ are linearly independent. □

3. $\begin{bmatrix} 1 \\ -3 \end{bmatrix}$, $\begin{bmatrix} -3 \\ 9 \end{bmatrix}$ $\boxed{\text{Solution:}}$ $\begin{bmatrix} -3 \\ 9 \end{bmatrix} = \begin{bmatrix} -3\cdot 1 \\ -3\cdot(-3) \end{bmatrix} = (-3)\cdot\begin{bmatrix} 1 \\ -3 \end{bmatrix}$ (Scalar multiplication)

$\begin{bmatrix} -3 \\ 9 \end{bmatrix}$ is a scalar multiple of $\begin{bmatrix} 1 \\ -3 \end{bmatrix}$.

∴ The given vectors are linearly dependent. □

In Exercises 5-8 determine if the columns of the matrix form a linearly independent set. Justify each answer.

5. $\begin{bmatrix} 0 & -8 & 5 \\ 3 & -7 & 4 \\ -1 & 5 & -4 \\ 1 & -3 & 2 \end{bmatrix}$. The columns of a matrix form a linearly independent set if and only if the matrix equation $Ax = 0$ has the trivial solution only.

By Theorem 3, the matrix equation $Ax = 0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.

$$\begin{bmatrix} 0 & -8 & 5 & 0 \\ 3 & -7 & 4 & 0 \\ -1 & 5 & -4 & 0 \\ 1 & -3 & 2 & 0 \end{bmatrix} \xrightarrow[\text{and Row 4}]{\text{Interchange Row 1}} \begin{bmatrix} 1 & -3 & 2 & 0 \\ 3 & -7 & 4 & 0 \\ -1 & 5 & -4 & 0 \\ 0 & -8 & 5 & 0 \end{bmatrix} \xrightarrow[-3\times Row1]{Row2 = Row2} \begin{bmatrix} 1 & -3 & 2 & 0 \\ 0 & 2 & -2 & 0 \\ -1 & 5 & -4 & 0 \\ 0 & -8 & 5 & 0 \end{bmatrix}$$

$$\xrightarrow[+Row1]{Row3 = Row3} \begin{bmatrix} 1 & -3 & 2 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & -8 & 5 & 0 \end{bmatrix} \xrightarrow[-Row2]{Row3 = Row3} \begin{bmatrix} 1 & -3 & 2 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -8 & 5 & 0 \end{bmatrix} \xrightarrow[\text{Row 3 and Row 4}]{\text{Interchange}}$$

$$\begin{bmatrix} 1 & -3 & 2 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & -8 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[+4\times Row2]{Row3 = Row3} \begin{bmatrix} 1 & -3 & 2 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\times(-1/3)]{Row3 = Row3} \begin{bmatrix} 1 & -3 & 2 & 0 \\ 0 & 2 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$